

SQL Saturday Austin Precon

Enabling AI in your Data Practice

Friday, June 26th, 2026

Presented by Ginger Grant

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Course Setup

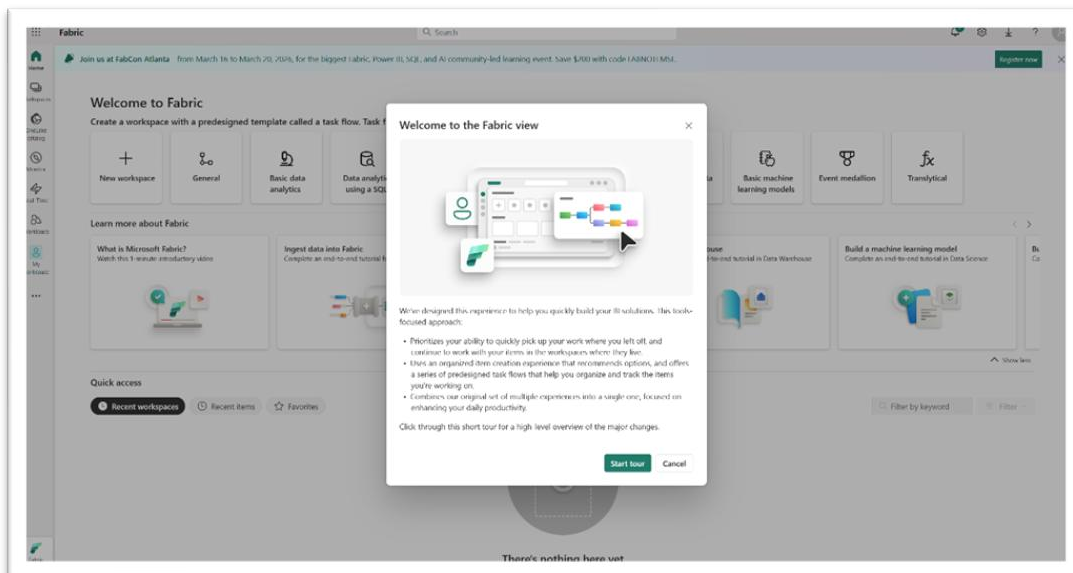
1. In your browser, go to <https://app.fabric.microsoft.com/>



2. Login with the email address User#.Fabric@desertislesql.com

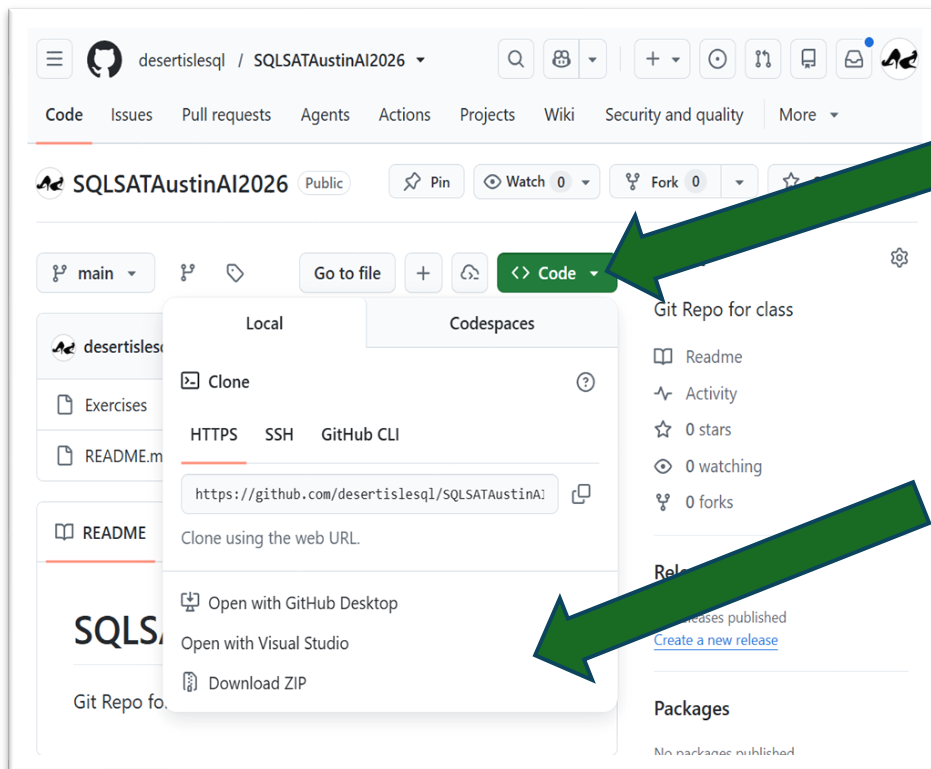
Password is **tH3AI_P.55w0rd**

- use the assigned number



When you are successful you will see this screen. Congratulations! You are signed in and ready to work with Fabric

3. Download the contents of the github repo
<https://github.com/desertislesql/SQLSATAustinAI2026>



4. Click on the green **Code** button when you do you will see a popup menu.
5. Select the option to **Download Zip** Remember where you save and unzip this as we will be using it in class
6. For class you will also need to use **SQL Server 2025 Enterprise Developer** and **SQL Server Management Studio 22**. If you do not have both of these installed for SQL Server link on the **SQL Server 2025 Developer** icon <https://www.microsoft.com/en-us/sql-server/sql-server-downloads?msocid=2aa56919c69361a51bf07ffec7ed6016> Download SQL Server Management Studio 22 from <https://learn.microsoft.com/en-us/ssms/install/install>

Prompt Engineering Test Data

1. Check to see if AdventureWorksDW2020 is available on SQLServer 2025 by using SQL Server Management Studio 22. If you do not have both of these applications installed, install them now.
2. If you have AdventureWorksDW2020 loaded, go to step 3. If not complete the following instructions.
 - A. Open **SSMS**
 - B. Connect to your SQL Server 2025 instance
 - C. Right-click **Databases**
 - D. Select **Restore Database...**
Source = Device
Click ... button
 - E. Click **Add**
 - F. Browse to the location where you downloaded the class repo and go to the Data folder and Select AWDW2020.bak
 - G. Click **OK**
 - H. In **Destination database**, enter: AdventureWorksDW2020
 - I. Fix file paths by going to the **Files** page (left side) where you will see logical files like:
AdventureWorksDW202_Data
AdventureWorksDW202_Log
 - J. Update the **Restore As** paths if needed, for example:
 - K. C:\Program Files\Microsoft SQL
Server\MSSQLXX.MSSQLSERVER\MSSQL\DATA\AdventureWorksDW2020.mdf
 - L. C:\Program Files\Microsoft SQL
Server\MSSQLXX.MSSQLSERVER\MSSQL\DATA\AdventureWorksDW2020_log.ldf
Click **OK**
You should see: Database 'AdventureWorksDW2020' restored successfully.
3. Open SQL Server Management Studio 22 (yes the version matters)
4. Connect to localhost and open up AdventureWorksDW2020
5. Go to FactInternetSales and right click on the table and Script Table **as Create To** and then select **New Query Editor Window**
6. Copy the Create Table statement.
7. Open your AI tool of choice (Claude Desktop, copilot.microsoft.com, Microsoft Edge Copilot pane or go to the view menu of SSMS and select GitHub Copilot Chat)
8. In the prompt window add this
Create a script to add data to this table so that it has order data up to 6/24/2026 and make sure that you use the key values found in the database for DimProduct

Paste in in the create table statement. Execute the prompt. Note if you are in GitHub Copilot Chat you can insert the script at the Cursor and press the Keep button. If you do that make sure you don't run the create table statement again. If you get a message just press **Confirm**

9. Run the script

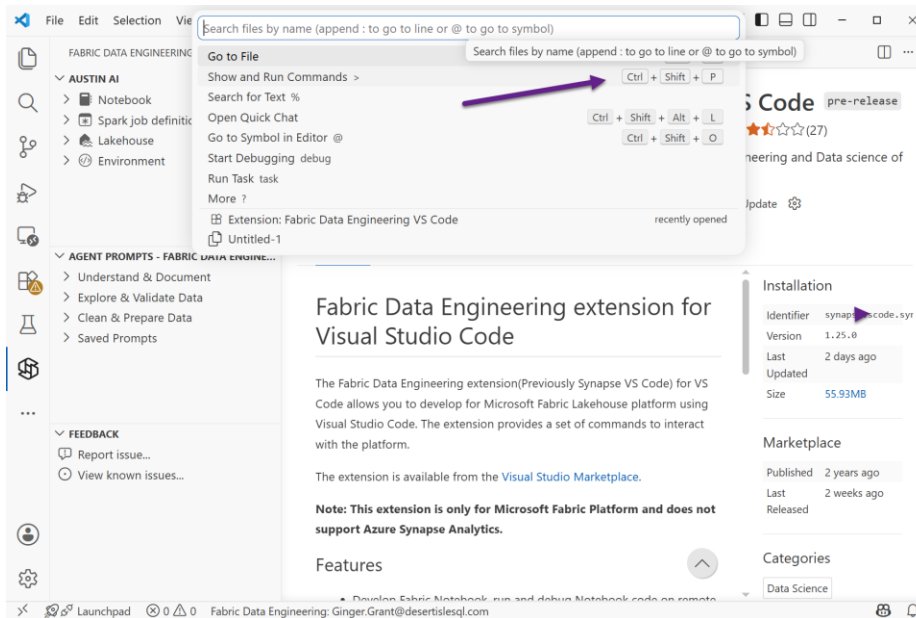
10. Create a prompt to check to make sure the data was added correctly.

Create a sql statement on table dbo.FactInternetSales showing the year of the orderdate

Run the sql you get back to see what was added. Think about doing this next time you need test data. You can have fun with it and include names of muppet characters as people.

VS Code Fabric Data Engineering with GitHub Copilot

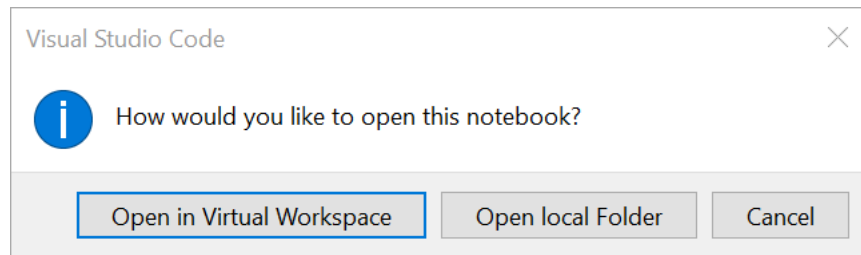
1. In your Browser, goto to app.fabric.Microsoft.com
2. Go to the Workspace Austin AI
3. Create a folder where you can store all the work for the class using either your name or your initials.
4. Open up VSCode and select File->New Window
5. In the new window, select the extension icon and check to see that you have Fabric Data Engineering installed
6. Click on the search bar at the top of the screen and Select **Ctrl+Shift+P**



7. Select Fabric Data Engineering: Select Workspace
8. From the list select **Austin AI**
9. Go to the view menu and select Chat to open the GitHub.
10. Modify the prompt to substitute your folder and initials Type the prompt In the Austin AI workspace in the folder <your folder> Create a lakehouse called <yourInitials>_lkData
11. Go to the Fabric service and open up the lakehouse you created
12. Create a new table based on a shortcut from XXX_OpDataStore. In the lakehouse explorer, click on the tables folder and from the magical ellipse menu select **New shortcut**
13. Select Microsoft OneLake
14. Select XXX_OpDataStore lakehouse and click the **Next** button.
15. Access the table: dbo.Hawkeye_liquor
16. Click on the **Create** button to complete the shortcut
17. Go back to VSCode and add this new prompt substituting XXX with your initials

Create a new notebook in the workspace Austin AI called XXX_nbData. Add two cells to the notebook. The first should be a configure statement to add XXX_lkData as the default lakehouse and create a select statement to get the categoryname, city and the total Saleamount from hawkeyeliquor

If you see this window



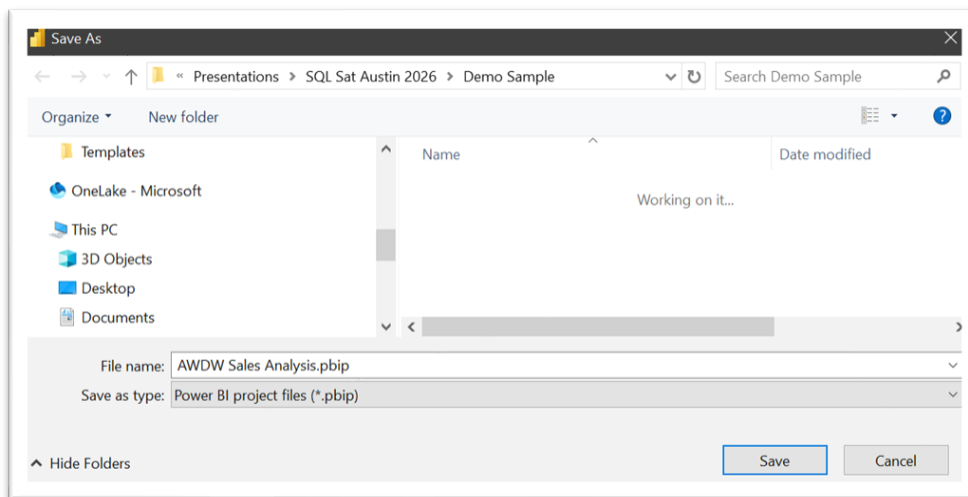
Select Open in Virtual workspace

18.Run the notebook in VSCode and then go back to the service and do the same.

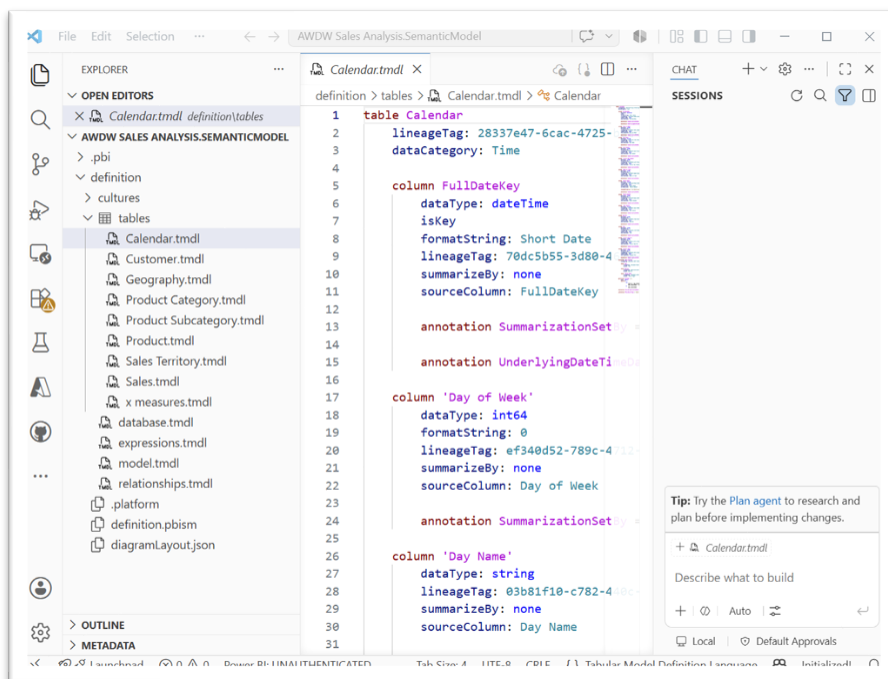
End of Exercise

Using GitHub Copilot with Power BI

1. Using the data downloaded from GitHubRepo, Open the PBIX file AWDW Sales Analysis.pbix
2. Go to the Model view by clicking on the model icon on the right side of the screen.
3. In the data view, click on the different tables and look at the description in the Properties Pane, look at all of the descriptions (spoiler alert, there aren't any)
4. Click on the x measures table and click on each of the DAX Measures.
5. In the data view, click on the different measure and look at the description in the Properties Pane for each DAX Measure, look at all of the descriptions (spoiler alert, there aren't any)
6. Go to File, Save as, browse to some location, and change the Save as Type to PBIP file.



7. Click on the Save Button.
8. Close Power BI.
9. Open VSCode. Make sure you have the TMDL extension installed. Click on the extension and the folder AWDW Sales Analysis.SemanticModel.
10. Open up the tables folder, then open up Calendar.tmdl. Your screen should look like the one below.



11. Open up the Copilot pane and enter this prompt

Tables in Copilot can be commented by putting /// above the table name. Provide appropriate descriptions for each table

Make sure that the agent mode is selected. It looks like this



12. Click on the send arrow

13. The copilot pane will fill up with things you should read. Tell it to keep the changes by clicking on the **Keep** button in the copilot pane if you want to click on it once. If you click on the keep button in the code pane you will be taken to each change which means you will have to click on it 8 times, one for each table.

14. Highlight the table x.Measures and enter the following prompt in the Github Copilot Chat

Add descriptions for all of the DAX measures

Click on the send arrow

15. Read all of the stuff in Github copilot and then click on the keep button.

16. Open up the report AWDW Sales Analysis.pbip in Power BI. Review changes and close.

End of Exercise

GitHub in SQL Server 2025

1. Open SSMS 22 and open up AdventureworksDW2020
2. In copilot chat enter the following
- 3.

Generate a query against AdventureWorksDW2020 that shows:

Calendar Year
Sales Territory
Product Category

Internet Sales Amount

Aggregate by year, territory, and category.

4. Review generated SQL. Execute. Show results.
5. Create a query with this SQL

```
SELECT
    d.CalendarYear,
    SUM(f.SalesAmount) AS SalesAmount
FROM FactInternetSales f
INNER JOIN DimDate d
    ON f.OrderDate = d.FullDateAlternateKey
GROUP BY d.CalendarYear
```

6. In Github copilot chat Type
Explain this query as if I were a business analyst.
7. Optimize a Bad query

```
SELECT *
FROM FactInternetSales
WHERE YEAR(OrderDate) = 2020
```

8. In Github copilot chat Type

Optimize this query for performance.

Examine the explanation and see what it suggests

9. Build a stored procedure that returns Internet Sales by Product Category for a selected year.
In Github copilot chat type
Create a stored procedure for AdventureWorksDW2020 that accepts a year parameter and returns sales by category.
10. Execute the stored procedure and test it

End of Exercise

Fabric MCP Server

Make sure you install the Fabric MCP Server prior to completing this exercise. In VSCode.

1. In VSCode type Ctrl+Shift+P and select MCP: List Servers
2. If the Fabric MCP Server is not running start it

3. Go to the GitHub Copilot Pane making sure you are in Agent mode. Type this in the window

What Fabric workload types are available?

Review the results

4. Type this in the window

List my Microsoft Fabric workspaces and find the workspace named Austin AI

5. Inspect the Lakehouse

In the Austin AI workspace, inspect the lakehouse named XXX_lkData List the available tables, folders, and files.

Ask for a cleaner summary:

Create a short inventory report for the XXX_lkData lakehouse showing tables, files, and folders.

6. Let's start documenting the Lakehouse

Using the metadata from the XXX_lkData lakehouse, create documentation that includes:

- Lakehouse purpose
- List of tables
- Probable business meaning of each table
- Key columns
- Potential data quality checks
- Recommended owners or subject areas

This will create the SQL Validation queries

7. Prompt copilot

Create a Fabric notebook plan that profiles all tables in the XXX_OpDataStore lakehouse and writes the results to a data quality summary table

Expected output will provide you with a plan containing:

- Connect to the lakehouse.
- Loop through lakehouse tables.
- Capture row counts.
- Capture null counts.
- Capture duplicate counts.
- Save results to a summary table.
- Produce a final status report.

End of Exercise

Fabric CLI

For this demo we are going to migrate a notebook from Austin AI test workspace to Austin AI Test workspace.

1. Go to the command prompt by typing CMD in the Search window
2. Check to see if you have python installed in the path

```
python --version
```

3. Install Fabric CLI

```
pip install ms-fabric-cli
```

4. Validate the install

```
Fab -version
```

5. Sign in to Fabric. When prompted, select Interactive with a web browser
6. Once you are logged in list all the items in the Austin AI workspace

```
fab ls "Austin AI" -l
```

7. Export the notebook definition.

```
fab export notebook `
    "Customer Sentiment Analysis" `
    --workspace "Austin AI" `
    --path .\NotebookExport
```

8. Review the Notebook with AI by opening the notebook in VS Code. Add this prompt in GitHub Copilot:

Review this notebook and identify:

- *Performance improvements*
- *Documentation gaps*
- *Security concerns*
- *Code quality issues*

Review the results to see how you can improve the code.

9. Go back to the command prompt and type

```
fab ls "Austin AI Test" -l
```

10. Deploy the Notebook and Import into the test workspace.

```
fab import notebook `
  --workspace "Austin AI Test" `
  --path .\NotebookExport
```

11. Verify Deployment List notebooks in the target workspace.

```
fab ls "Austin AI Test" -l
```

While you can do this in the UI, Fabric CLI provides another way to move notebooks. Instead of manually recreating notebooks in another workspace, Fabric CLI allows us to treat notebooks like code.

End of Exercise

GINGER GRANT

You're collecting **gobs of data**.

Every department is **doing their thing**.

You're making **data-driven decisions**...
inside the silos.

Isn't this enough? Not exactly.

TRUTH: Data is NOT driving you forward if you don't have a one system with an enterprise-level view: A single foundation that people understand, embrace and rally around. But it feels way too big to tackle.

You want to...

- Implement new technology
- Stay with the Microsoft Data Stack
- Train your employees to use all of it.

But you may not know how. I understand and I can help.

THE OUTCOMES

- ✓ A custom architecture designed to scale
- ✓ Positioned for aligned investment in machine learning & AI
- ✓ Shared training & commitment to data quality
- ✓ Predictive analytics to help you plan
- ✓ Confident, data-driven decisions
- ✓ Data you can trust

Let's Connect!



<https://www.linkedin.com/in/ginger-grant/>



ABOUT GINGER

Data Consultant since 2004

Microsoft Data Platform MVP since 2016,
(384 worldwide)

Deep Understanding
of Data Architecture

Vast Network
of Resources

I Teach & Translate
All Things Tech



I Code Custom Generative AI Solutions



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